```
Andrii Zeliuk, IT-41, N = 11
--- Task 1 ---
[ 1  2  3  4  5  6  7  8  9 10 11]
shape: (11,)
size:  11
dtype: int64
--- Task 2 ---
[[0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]]
[[1 1 1 1 1 1 1 1 1 1 1]
 [1 1 1 1 1 1 1 1 1 1 1]]
[[11 11]
 [11 11]]
[[1. 0. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 1. 0. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 1. 0. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 1. 0. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 1. 0. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 1. 0. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 1. 0. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 1. 0. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 1. 0.]
 [0. 0. 0. 0. 0. 0. 0. 0. 0. 1.]]
--- Task 3 ---
[0.  0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1. ]
[ 0  2  4  6  8 10 12 14 16 18 20]
```

```
--- Task 4 ---
[ 1  2  3  4  5  6  7  8  9 10 11 12]
[[ 1  2  3  4]
 [ 5  6  7  8]
 [ 9 10 11 12]]
(3, 4)
[[ 1  2  3]
 [ 4  5  6]
 [ 7  8  9]
 [10 11 12]]
(4, 3)
[[ 1  2  3  4  5  6]
 [ 7  8  9 10 11 12]]
(2, 6)
--- Task 5 ---
[ 2  4  6  8 10 12 14 16 18 20 22]
[11 12 13 14 15 16 17 18 19 20 21]
[  1   4   9  16  25  36  49  64  81 100 121]
[0.09090909 0.18181818 0.27272727 0.36363636 0.45454545 0.54545455
 0.63636364 0.72727273 0.81818182 0.90909091 1.        ]
506
--- Task 6 ---
8.0
15.5
11.428571428571429
4
[False  True False  True  True  True False]
4
[50.9 53.6 49.1 55.4 59.9 52.7 46.4]
```

```
--- Task 7 ---
78
[15 18 21 24]
[10 26 42]
```